BANGLADESH ARMY UNIVERSITY OF ENGINEERING & TECHNOLOGY (BAUET)

AUDIO-BASED DISTRESS AND FALL DETECTION SYSTEM



Course Code: CSE-3200 | Software Development Project (Phase-I)

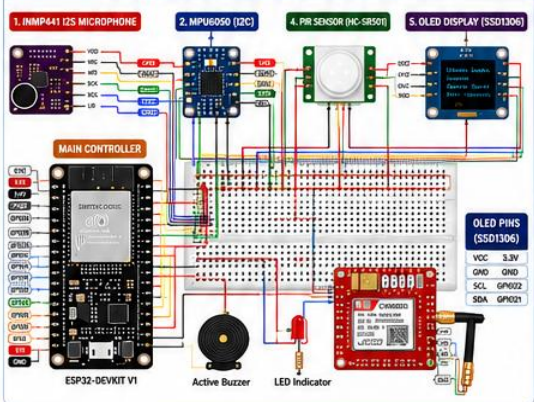
1. INTRODUCTION

Elderly individuals, especially those living alone, are at risk of accidents like falls or medical emergencies. Traditional monitoring methods are often intrusive or impractical. Our system uses audio analysis to detect distress and fall events from environmental sounds and generates instant alerts to caregivers. It is a low-power, non-intrusive and privacy-preserving solution for smart elderly care.

2. PROJECT OBJECTIVES

- Develop a low-power embedded audio monitoring device.
- Capture and analyze environmental audio in real time.
- Detect fall and distress (help, pain, emergency) sounds.
- Send instant alerts to caregivers via SMS / App.
- Provide a non-intrusive and privacy-friendly solution.

CIRCUIT CONNECTION DIAGRAM



3. DATASET

SAFE DATASET

The SAFE dataset contains realistic audio recordings of daily activities and ambient sounds for research purposes.

- Falls
- Walking
- Talking
- Daily Activities
- Environmental Sounds

Source: Kaggle – SAFE Dataset

CUSTOM DATASET

Custom recordings collected for distress and emergency related sounds.

- Help
- Pain
- Distress
- Emergency Voice
- Real Recorded Sounds

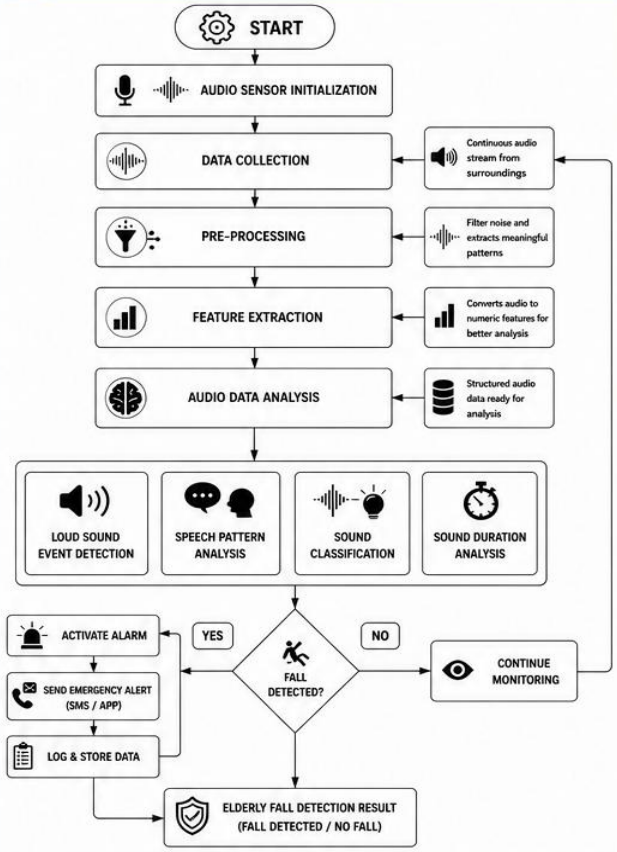
Collected in Real Environment

SYSTEM WORKFLOW

Microphone (Audio Capture) → Audio Collection → Noise Removal → Feature Extraction → Rule-Based Decision → Alarm Trigger → SMS / App Notification

All processing is done on the local device (edge) for faster response and privacy.

4. METHODOLOGY



5. FUTURE WORK

- Improve accuracy using advanced signal processing techniques.
- Add multi-sensor support (motion, temperature, heart rate).
- Develop a mobile application for real-time monitoring.
- Implement cloud storage for long-term data access.
- Extend system for multi-room and outdoor environments.
- Integration with smart home and IoT platforms.



6. CONCLUSION

This project presents a rule-based audio processing system that can detect falls and distress events using environmental audio. By utilizing meaningful audio features and threshold-based decision logic, the system ensures real-time monitoring, immediate alerts and improved safety for elderly individuals while maintaining privacy and independence.

AUDIO MONITORING — Continuous monitoring of ambient sounds

EDGE PROCESSING — On-device processing for fast response

REAL-TIME ALERT — Instant alerts to caregivers via SMS / App

SAFE & NON-INTRUSIVE — Privacy-friendly and respectful monitoring

TEAM MEMBERS

1 Md Jyaul Islam Jibon — ID : 0812310205101009
2 Jannatul Ferdous Mou — ID : 0812310105101011
3 Maksura Mahjabin — ID : 0812310105101022

SUPERVISOR
Subrata Kumar Paul
Assistant Professor
Department of CSE, BAUET